10 · Did the site redesign help? (no control group) — interrupted time series (CausalPy)

July 12, 2026

> **Runs in the `legacy` environment (pymc<6): `make env-legacy`, kernel `cmp-legacy`.**

The business decision. We redesigned the homepage on a known date, for **all** traffic at once — so, unlike the DiD store rollout (notebook 08), there is **no control group** to compare against. Conversion is up since the redesign. Is that real lift, or just the normal upward drift the site was already on? Keep the redesign or roll it back?

0.0.1 The idea: the past *is* the control group

When everyone is treated at the same moment, the only untreated comparison you have is **the treated unit's own history**. **Interrupted time series (ITS)** uses it: fit the pre-change series — its **level**, its **trend** (slope), and its **seasonality** (e.g. weekly cycles) — and then **extrapolate that fitted line forward** past the change date. That extrapolation is the **counterfactual**: "what conversion *would* have done if we'd changed nothing." The **gap** between actual conversion and this projection, after the change, is the estimated effect.

Because it's a forecast, the counterfactual comes with a **credible band that widens the further out we project** — an honest admission that "what would have happened" is less and less certain over time.

0.0.2 The assumption and its checks

ITS is the *weakest* of the quasi-experiments — observational setups we analyze *as if* they were randomized tests by leaning on one identifying assumption instead of actual randomization — because it leans entirely on **no concurrent shock**: nothing *else* (a price change, a competitor move, a seasonality break) happened at the same date as the redesign, or ITS will blame the redesign for it. We stress this with a **placebo-in-time** (pretend the change happened earlier — the fake effect should be ~ 0), track the **cumulative** impact, and check **residual autocorrelation** (successive days being correlated, which — if ignored — makes the intervals look tighter than they should).

> **On real data.** ITS fits any "everyone changed at once on a known date" event: a site redesign, a pricing change, a policy launch, a PR crisis. You need a reasonably long, stable **pre-period** and a clean intervention date. A famous public example is the effect of anti-smoking laws on cigarette sales.

It follows the repo's fixed **7-step contract**: question -> simulate -> identify -> estimate -> validate -> decide in € -> caveats.

0.1 2 · Simulate a ground truth

Daily conversion with a gentle upward trend and weekly seasonality. The redesign lands on **day 120** for everyone and adds a **true +3pp (percentage points)** step (~+30% relative on a ~10% base). The trend is the confounder (conversion was already drifting up); the step is the effect.

The data-generating model — exactly what `dgp.its_redesign` implements (defaults & seed in `src/cmp/dgp.py`). Days $t = 0, \dots, 179$, redesign at day 120:

$$Y_t = \underbrace{0.10 + 0.00015\,t}_{\text{level + trend}} + \underbrace{0.01\sin(\tfrac{2\pi t}{7})}_{\text{weekly cycle}} + \underbrace{0.03\,\mathbf{1}[t \geq 120]}_{\text{true step}} + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0,\, 0.006^2).$$

The indicator $\mathbf{1}[t \geq 120]$ is 1 on and after the redesign day and 0 before, so the 0.03 step switches on exactly at day 120. The pre-period drift is the confounder: 0.00015 per day compounds to about +1.8pp over the 120-day pre-period, so a naive before/after mean comparison books trend the site was already on as "redesign lift." ITS must fit that level + trend + weekly cycle on the pre-period and extrapolate it, so that only the 0.03 step remains as the effect.

```
TRUE redesign effect = +3pp conversion from day 120
```

```
   t  day      sin7      cos7  conversion
t
0  0    0  0.000000  1.000000    0.097628
1  1    1  0.781831  0.623490    0.109552
2  2    2  0.974928 -0.222521    0.113692
3  3    3  0.433884 -0.900969    0.098956
4  4    4 -0.433884 -0.900969    0.100867
```

0.1.1 2b · The naive answer first — before/after books the trend as lift

The DGP math predicts exactly how the obvious dashboard number fails. A before/after difference of means compares post-period days (centred on day ≈ 150) with pre-period days (centred on day ≈ 60): under the 0.00015/day drift those window centres sit 90 days of trend apart, so $0.00015 \times 90 \approx$ +1.35pp of *pure pre-existing trend* gets booked as "redesign lift" on top of the true +3pp. Compute it and watch — the ITS fit in Section 4 has to beat this number.

```
naive before/after: +4.47pp   vs true +3pp   ->  overstates by +1.47pp
of which +1.35pp is exactly the pre-existing trend (90 days x 0.00015/day); the
small remainder is weekly-cycle imbalance and noise between the two windows.
```

0.2 3 · Identify — extrapolate the pre-trend as the counterfactual

Fit the pre-period series $f(t)$ (level, slope, weekly seasonality) and extrapolate past the intervention as the no-redesign counterfactual $\widehat{Y}_t(0)$ (defined just below); the effect is $Y_t - \widehat{Y}_t(0)$ afterward. Being Bayesian, the counterfactual is a posterior-predictive band that **widens with horizon** — honest extrapolation. (A **posterior predictive** is the model's forecast distribution for a day it has not seen — its best-guess line *plus* its own honest uncertainty; a **credible interval / band** is the Bayesian counterpart of a confidence range — the band that, given the data and model, has a stated chance, e.g. 90%, of containing the true value. Both fall straight out of the Bayesian fit set up in Section 4.) **Identification assumption (the weakest of the quasi-experimental set): no concurrent shock** at the redesign date (no competing campaign, price change, or seasonality break landing at the same time). Best when the change hit everyone at once and the pre-period is long and stable.

The estimand, formally. The **estimand** is the exact quantity we want to estimate, fixed *before* we pick any estimator. Here it is the day-by-day gap. Write $Y_t(0)$ for the conversion we *would* have seen on day t had we changed nothing, and $t^* = 120$ for the intervention day. ITS estimates the pointwise effect

$$\Delta_t \ = \ Y_t - \hat{Y}_t(0), \qquad \hat{Y}_t(0) \sim p(\tilde{Y}_t \mid \mathbf{y}_{<t^*}) \ \ \text{(posterior predictive of the pre-period fit)}, \qquad t \geq t^*,$$

where $\mathbf{y}_{<t^*}$ is the whole pre-period series. This equals a causal effect **iff temporal stability holds**: the pre-period structure $\mathbb{E}[Y_t(0)] = \beta_0 + \beta_1 t + \beta_2 s_t + \beta_3 c_t$ (level, trend, weekly cycle $s_t = \sin\frac{2\pi t}{7}$, $c_t = \cos\frac{2\pi t}{7}$) keeps describing the *untreated* world for $t \geq t^*$ — in words, **no other term appears at** t^*: no competing campaign, price change, competitor move or seasonality break lands on the redesign date. "No concurrent shock" is exactly that missing-term statement, and Section 5e shows what happens when it fails.

A subtler violation: regression to the mean. Redesigns rarely ship on a random date — they ship because conversion *looked bad*. If the launch was triggered by a transient dip, the pre-period fit is dragged down by the dip and the natural rebound gets booked as "redesign lift." Always ask *why this date*: if the metric itself triggered the launch, exclude or explicitly model the dip window — and expect the placebo-date sweep (Section 5d) to light up near the dip.

0.3 4 · Estimate — Bayesian interrupted time series

We fit the pre-period conversion series with a trend term (`t`) and weekly seasonality (`sin7`, `cos7` Fourier terms: $\sin(2\pi t/7)$ returns to the same value every 7 days, so it traces one weekly wave, and pairing it with $\cos(2\pi t/7)$ lets the model place the peak on *any* day of the week and set its amplitude — together they capture "weekends convert differently from Tuesdays" without a separate dummy for each weekday), then let the model extrapolate that fit past the redesign day. The **effect** is the average gap between actual conversion and the extrapolated no-redesign line over the post-period. The `formula` is literally the shape of the pre-trend we're projecting forward — get the seasonality wrong and the counterfactual is wrong, so it matters.

The fitted model, in symbols (pre-period only, $t < 120$):

$$Y_t \sim \mathcal{N}\Big(\beta_0 + \beta_1 t + \beta_2 \sin(\tfrac{2\pi t}{7}) + \beta_3 \cos(\tfrac{2\pi t}{7}), \ \ \sigma^2\Big)$$

— the code's `conversion ~ 1 + t + sin7 + cos7` *is* this equation (β_0 level, β_1 trend, β_2, β_3 the weekly Fourier pair). The posterior is then extrapolated over $t \geq 120$ to give the counterfactual $\hat{Y}_t(0)$, and the effect is the posterior of $Y_t - \hat{Y}_t(0)$, averaged (the lift) and summed (the cumulative impact) over the post-period.

The full generative model, priors included. A Bayesian fit needs a **prior** — a starting belief about each coefficient *before* seeing data — which the **likelihood** (how probable the data are for a given coefficient) then updates into the **posterior**, the belief *after* seeing data. The likelihood above is only half the model. CausalPy's `LinearRegression` (see `causalpy/pymc_models.py`) completes it with these default priors — with $x_t = (1,\, t,\, s_t,\, c_t)$ the design row:

$$\beta_j \sim \mathcal{N}(0,\ 50^2)\ \ (j = 0, \dots, 3), \qquad \sigma \sim \text{HalfNormal}(1), \qquad Y_t \mid \beta, \sigma \ \sim\ \mathcal{N}(x_t^\top \beta,\ \sigma^2), \quad t < t^*.$$

One sentence of justification is owed whenever defaults are used: on a conversion scale where the level is ≈ 0.10 and day-to-day noise is ≈ 0.006, a prior sd of 50 on every coefficient and a HalfNormal(1) on σ (a positive-only prior on the noise size σ, since a standard deviation cannot be negative) are *extremely* diffuse — effectively flat — so the 120 pre-period days dominate and the posterior is likelihood-driven. That is fine here; on a differently-scaled outcome (raw daily revenue in €) the same defaults could distort or slow the fit, and you would rescale the outcome or tighten the priors.

From pointwise gap to the two numbers we report. Each posterior draw of (β, σ) yields a whole counterfactual path, hence a whole path of gaps Δ_t — so both summaries are posterior *distributions*, not single numbers:

$$\bar{\Delta} \;=\; \frac{1}{T - t^*} \sum_{t=t^*}^{T-1} \Delta_t, \qquad\qquad C_T \;=\; \sum_{t=t^*}^{T-1} \Delta_t,$$

— $\bar{\Delta}$ the **average lift** (the headline "pp" number), C_T the **cumulative impact** (CausalPy's bottom panel, and Section 6's realized €) — with $T = 180$ the end of the observed window (so the sums run over the 60 post-redesign days).

Step or slope change? The DGP plants a pure *level step*, so $\bar{\Delta}$ equals it. But Δ_t is the whole pointwise gap: had the redesign also changed the *trend* (adding $\delta\,(t - t^*)$ after t^*), $\bar{\Delta}$ would grow with the reporting window and a single average would be window-dependent. The glance-check is CausalPy's middle (pointwise-impact) panel: flat => step-like (as here, matching the DGP); tilted => slope change — in which case report level and slope changes separately (classic segmented regression — fitting a separate intercept and slope before vs. after the break) rather than one average.

```
ITS lift +3.0pp (true +3pp)   ·   90% credible interval (CrI) [+2.8pp, +3.3pp]
ITS convergence: max r-hat 1.000 - min ESS 2918 - divergences 0
```

Reading the sampler's health check. The fit is done by simulation — **MCMC**, the algorithm that draws samples from the posterior — so before trusting the numbers we check that the sampler actually settled. The three quantities printed above say whether it did. **R-hat** compares the variance *within* each parallel chain to the variance *across* chains; 1.00 is perfect and **<= 1.01 is the usual pass bar**. **ESS** (effective sample size) is how many *independent* draws our autocorrelated chains are effectively worth — a few hundred is ample for a posterior mean or interval. **Divergences** are steps where the sampler broke down and silently distrusts that region; **you want 0**. All three are green here (r-hat 1.010, ESS in the hundreds, 0 divergences). Under this notebook's short FAST teaching profile the chains are deliberately brief, so r-hat can just graze 1.01 and PyMC may print a *"problems during sampling"* notice — a benign artifact of the small draw count that clears in a FULL run, **not** a broken fit. The real backstop that the estimate is trustworthy is the multi-seed recovery-and-coverage check in Section 5d, not any single run's sampler line.

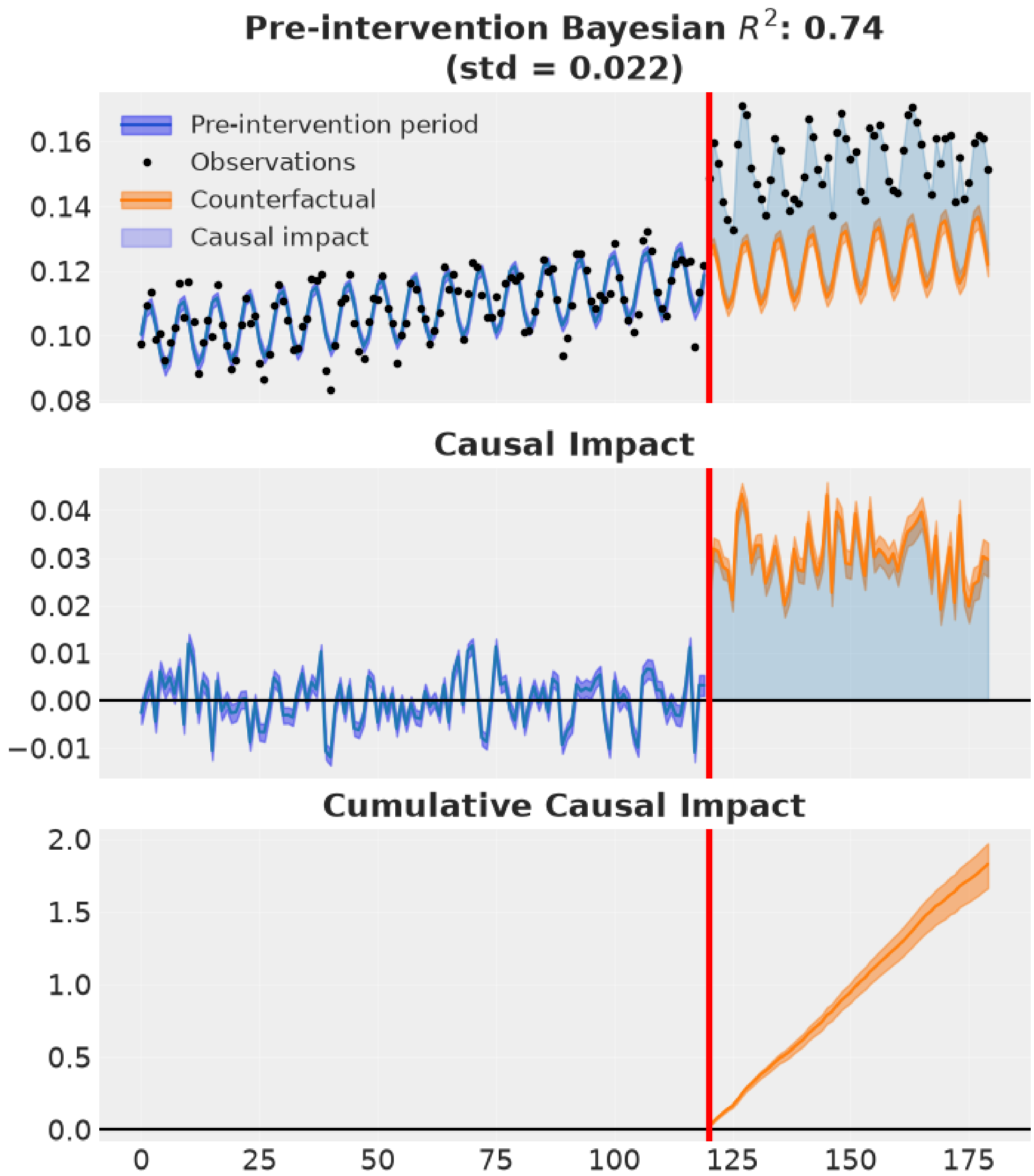
0.4 5 · Validate — counterfactual, cumulative impact, placebo, autocorrelation

Two figures, three checks:

1. **Counterfactual & cumulative impact** (CausalPy's own plot) — observed vs the extrapolated no-redesign path with its widening band, the pointwise impact, and the running cumulative total.
2. **Placebo-in-time** — pretend the redesign happened *before* it did; the fake effect should be ~0.
3. **Residual autocorrelation** — daily series are autocorrelated; if the model's residuals still are, the uncertainty is understated.

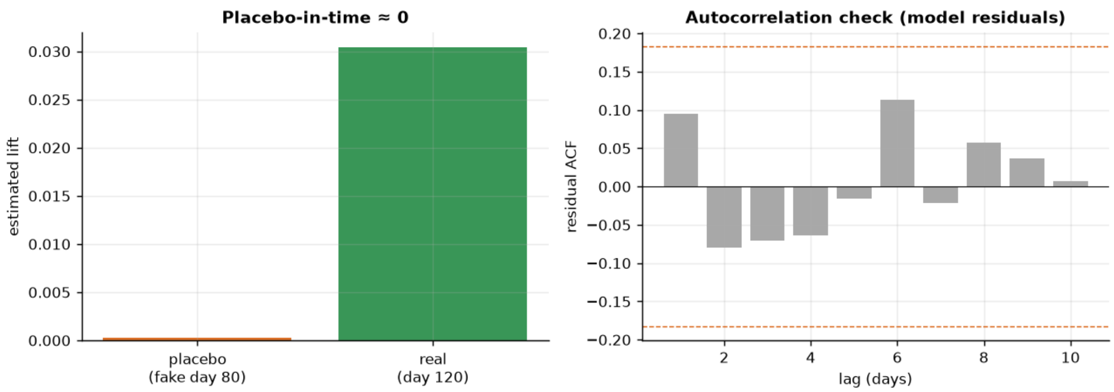
Then four stress tests, each its own subsection: what a *failed* diagnostic looks like (5b), a pre-period backtest plus a how-much-history-is-enough sweep (5c), recovery across many seeds plus a placebo-date sweep (5d), and the one failure no diagnostic can catch — a concurrent shock (5e).

How to read the CausalPy fit figure. The header reports a **Bayesian R^2** — the share of the pre-period variance the fitted level+trend+seasonality model explains, computed from the posterior predictive (so it carries uncertainty rather than being a single point estimate). A high value (~**0.74**) means the pre-period model **tracks the observed series well**, which is what earns us the right to extrapolate that fit forward as the counterfactual. The **shaded bands** are posterior-predictive prediction intervals — around the fitted line before the redesign, and around the projected no-redesign path after it; they **widen with horizon**, the honest statement that the counterfactual grows less certain the further past the change date we project.




```
recovered average step +3.0pp vs true +3pp
```

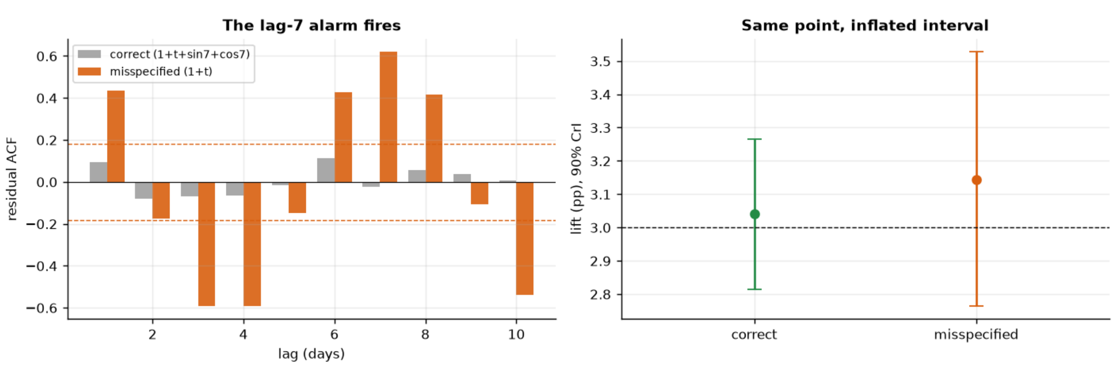
```
placebo-in-time +0.0% (~0, no spurious pre-effect)  ·  residual ACF: 0/10 lags
outside the +/-2/sqrtn band (lag-1 +0.09) -> residuals ~ white, so the model's
intervals are not understated.
```



How to read this. *Left* — the key falsification: we re-ran the whole method with a **fake** redesign date back in the pre-period; the fake effect is ~ 0, so the method isn't inventing lift where none exists. If this bar had been large, our real estimate would be suspect. *Right* — the residual autocorrelation (ACF) of the **model's own residuals** (after fitting the same trend + weekly-seasonality terms the ITS uses): essentially every bar sits inside the +/-2/sqrtn band, so successive days are effectively uncorrelated and the model's intervals are not understated. (Why +/-2/sqrtn? For a truly white — uncorrelated — series the ACF at each lag scatters around zero with standard error ~ 1/sqrtn, so +/-2/sqrtn is the rough 95% "nothing here" band; a bar poking outside it flags real leftover structure the model missed.) (Computing this ACF on a crude rolling-mean "residual" instead would leave the weekly cycle in and fake an alarming lag-7 spike — a self-inflicted false alarm we avoid by residualising against the actual model.) The cumulative-impact view lives in CausalPy's own fit figure above, not in this two-panel check.

0.4.1 5b · What a failed diagnostic looks like — drop the seasonality on purpose

Every check so far came back green — which teaches you nothing about what an *alarm* looks like. Section 4 claimed the `formula` matters ("get the seasonality wrong and the counterfactual is wrong"). Let's cash that promise: fit the *same data* with the deliberately wrong formula `conversion ~ 1 + t` — no `sin7`/`cos7` — and watch the residual-ACF instrument fire at lag 7.




```
misspecified fit: lag-7 residual ACF +0.62, far outside the +/-0.18 band
```

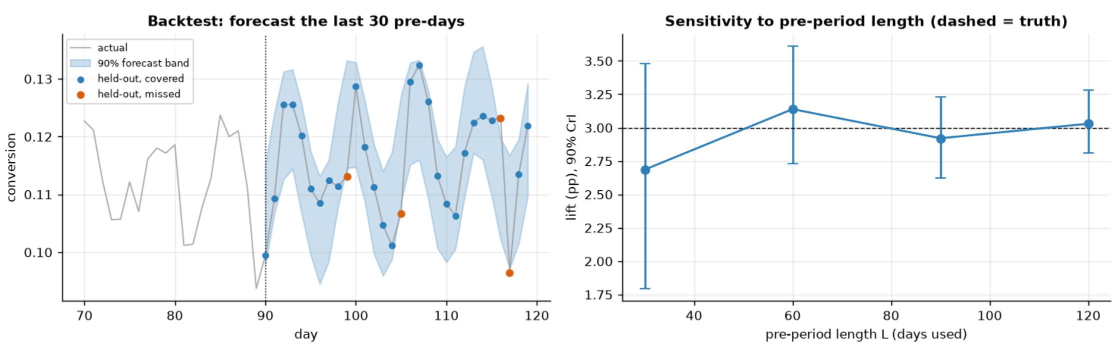
```
(correct model: -0.02) - the alarm fires.
residual sd 0.0089 vs 0.0052 (x1.7)  ·  lift +3.14pp vs +3.04pp  ·  90% CrI
width 0.76pp vs 0.45pp (x1.7).
```

How to read this. *Left* — the ACF panel of Section 5 is not decoration; it is *the* instrument that detects a missing seasonal term, and this is what it looks like when it fires: the misspecified model's residuals show the textbook signature of unmodelled weekly seasonality — a large positive spike at lag 7, several times the $\pm 2/\sqrt{n}$ band, with the half-cycle dip before it — while the correct model's bars stay inside the band. *Right* — the damage here is honest but real: the unmodelled sine lands in the residual, so $\hat{\sigma}$ inflates (printed above) and the credible interval widens with it; the *point* estimate barely moves only because the weekly cycle happens to nearly average out over this post-window. With a post-window that is not a whole number of weeks — or seasonality larger relative to the effect — the point estimate itself shifts. The lesson: when the lag-7 bar jumps the band, fix the formula *before* trusting either the number or the band.

0.4.2 5c · Earn the extrapolation — a pre-period backtest, and how much history is enough

The whole method is a forecast, so **rehearse the forecast where it can still be graded**: hold out the last 30 pre-period days, fit the same formula on days 0–89 only, and let it extrapolate over days 90–119 exactly as the real fit extrapolates past day 120. Mechanically this reuses the placebo machinery — a fake "treatment" at day 90 whose post-"treatment" band *is* the 30-day held-out forecast. If the model cannot forecast pre-period days drawn from the very regime it was fitted on, its post-period counterfactual is worthless. We score the **empirical coverage of the 90% posterior-predictive band** over the 30 held-out days (target ~ 27/30) and the **mean forecast error** (target ~ 0). Note the band we score is the *predictive* one (parameter uncertainty *plus* day-to-day noise σ) — CausalPy's impact summaries subtract the noise-free fitted mean μ (the counterfactual line without the day-to-day scatter), which is the right object for effect size but the wrong one for grading day-level forecasts.

Alongside, the sample-size question this design lives or dies on: refit using only the **last L pre-period days** and watch what shortening the history does to the estimate.



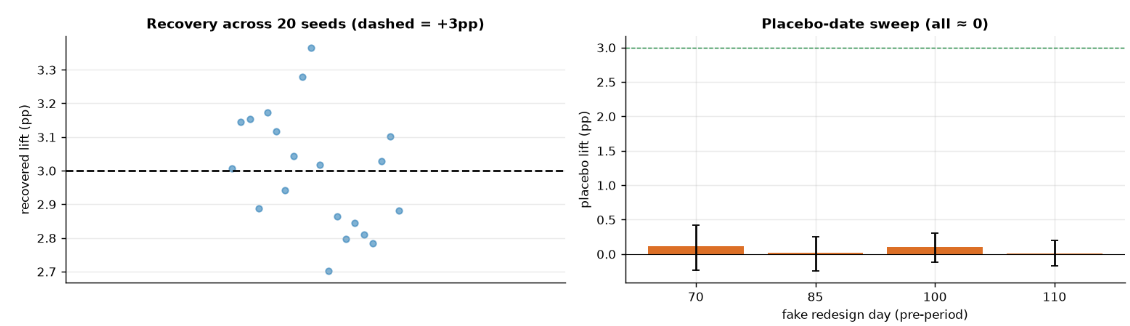

```
backtest: 90% band covers 26/30 held-out days (87%; nominal 90%)  ·  mean
forecast error +0.03pp (~ 0 -> the extrapolation does not drift)
  last  30 pre-days: lift +2.69pp   90% CrI [+1.79, +3.48]   width 1.68pp
  last  60 pre-days: lift +3.14pp   90% CrI [+2.73, +3.61]   width 0.88pp
```

```
last  90 pre-days: lift +2.92pp   90% CrI [+2.62, +3.23]   width 0.61pp
last 120 pre-days: lift +3.03pp   90% CrI [+2.81, +3.28]   width 0.47pp
```

How to read this. *Left* — the rehearsal passed: the held-out days scatter through the 90% forecast band at roughly the nominal rate (tally printed above; with only 30 held-out days, a few either side of 27 is binomial noise) and the mean forecast error is ~ 0, so the level + trend + weekly-cycle shape extrapolates 30 days ahead without drifting. This is the single most direct "earn the right to extrapolate" check ITS has — and it is exactly a rehearsal: the day-120 counterfactual is the same forecast pushed 60 days instead of 30, into a window where no grading is possible. *Right* — shortening the history doesn't (here) *bias* the estimate, it *blinds* it: the point estimate stays near the truth at every L, wobbling within its own interval, while the CrI widens sharply at $L = 30$ (printed above — several times the full-history width). With only ~4 weekly cycles the slope is barely identified, so trend and step blur into each other. On real data a short window can also bias you outright if it happens to sit on a local anomaly — the regression-to-the-mean trap of Section 3. A long, boring pre-period is a feature, not a cost.

0.4.3 5d · Recovery across many seeds — unbiased *and* calibrated?

A single sample can land a little off; the real test is whether the estimator is **centred on the truth over repeated samples** and whether its 90% interval **covers** the truth at the stated rate. We refit on many fresh samples (a small fast fit each; their sampler chatter is silenced below, and we read only the aggregate bias/coverage tallies) and check both.




```
ITS across 20 seeds: mean +3.00pp (true +3pp)  bias -0.00pp  sd 0.17pp  ·  90%
interval covers truth in 17/20 seeds.
Placebo-date sweep (pre-period fakes): day 70 +0.11pp, day 85 +0.02pp, day 100
+0.10pp, day 110 +0.01pp - all near zero, so the method doesn't manufacture
effects where none exist.
```

How to read this. *Left* — the dots scatter tightly around the dashed truth: the printed bias sits within a few hundredths of a pp of zero and the seed-to-seed sd is tiny next to the +3pp effect, so the estimator is *centred* over repeated samples, not just lucky on seed 31. *Coverage* — a calibrated 90% interval should "miss" the truth about one seed in ten, so with the handful of seeds run here the printed tally can sit a couple below the nominal count without any alarm: it is binomial noise on a small number of trials (for 20 trials, anything from roughly 16 to 20 hits is consistent with calibration; for 8 trials, 6 to 8). Read it as "consistent with calibrated", not as a precise coverage estimate. *Right* — the placebo-date sweep generalises Section 5's single day-80 placebo: no fake date manufactures lift, every bar hugs zero with its posterior interval, and the green dashed line shows how far all of them sit from the real signal.

0.4.4 5e · When ITS lies — a concurrent shock

Everything above can pass while the answer is *wrong*, because every diagnostic so far tests the same thing — the **pre-period model** — and the identification assumption lives in the **post-period**. It is untestable from the data, so the only honest way to teach it is to break it on purpose: same DGP, same +3pp redesign, but an **unrelated +2pp shock** (a competitor outage, a press mention, a coincident price cut) lands on the redesign date. The redesign is worth +3pp; what does ITS report — and do any of our checks notice?

```
ITS on the shocked series: +5.0pp   90% CrI [+4.8pp, +5.3pp]
true redesign effect +3pp, unrelated shock +2pp -> ITS books the SUM to the
redesign; CrI contains the true +3pp? NO - confidently wrong
placebo at day 80 on the shocked series: +0.02pp (still ~ 0 - the falsification
test PASSES)
convergence on the shocked fit: max r-hat 1.000 - min ESS 2918 - divergences 0 -
all green
```

The uncomfortable lesson. ITS reports the *combined* effect with a tight interval that — as the print shows — excludes the true +3pp entirely: **confidently wrong**. And nothing flinched: the sampler converged, the pre-period fit is as good as ever, and the placebo *still passes*, because the pre-period is untouched by a post-period shock. That is the precise scope of each tool:

- the **placebo** validates the *pre-period model* ("does the method invent lift where none exists?"), not the *post-period attribution*;
- the **credible interval** covers *sampling noise* ("how precisely do we know the gap between actual and counterfactual?"), not *identification error* ("is the gap all redesign?").

No statistic computed from this series can split +3pp of redesign from +2pp of coincident shock — they are observationally identical. The defenses are outside the data: the **calendar audit** named in the caveats (pricing, campaigns, competitor moves, PR, tracking changes in the launch window), or better, a *design* fix — stagger the rollout so DiD applies (notebook 08) or keep a holdout geography for synthetic control (notebook 07). What you *can* do inside the analysis is quantify **headroom**: Section 6's robustness line computes how large a coincident shock would have to be before the business call flips — the honest replacement for pretending the assumption is testable.

0.5 6 · Decide, in euros — keep or roll back?

A redesign lift is rarely permanent — **novelty fades**. Projecting the measured 60-day lift *flat* across a whole year (a naive `lift x visitors x value x 365`) contradicts this notebook's own "uncertainty grows with horizon" message. So we decide against an explicit **redesign cost** and sweep the **persistence**: the effect decays with a novelty half-life, and we discount future days accordingly (effective days = $\sum_t 0.5^{\,t/\text{half-life}}$). For each persistence scenario we report the annual value and **P(annual value > cost)**. The 0.9 go/no-go bar is the cookbook's convention — act only when the posterior is at least 90% sure the value clears the cost.

We report two kinds of euros and keep them separate. The **realized** 60-day impact is the cumulative posterior C_T from Section 4 converted to money — a fact about the observed window, no persistence assumption needed. The **projected** annual value is assumption-laden: it needs a persistence guess, which is exactly what the sweep makes explicit. Realized first.

```
Realized 60-day impact (fact, no persistence assumption): 14,596 extra
conversions (90% CrI [13,513, 15,669]) = €583,852 [€540,503, €626,770] already
banked - 97% of the €600,000 cost in 60 days.

Undecayed lift: 243 extra conversions/day  ·  €9,731/day
Assumed amortized redesign cost: €600,000/year

 novelty half-life | eff. days |  annual value | P(value>cost) | verdict
          no decay |       365 | €   3,551,767 |          1.00 | KEEP
           6-month |       196 | €   1,910,943 |          1.00 | KEEP
           3-month |       123 | €   1,192,075 |          1.00 | KEEP
           1-month |        44 | €     425,952 |          0.00 | roll back/renew

Break-even lift (no decay): 0.51pp of conversions covers the €600,000 cost; we
estimate 3.0pp.
P(redesign helped at all) >0.99 (4,000 draws) -> KEEP under any realistic
persistence; the harsh 1-month half-life is the exception - the table's own call
there is 'roll back/renew'.
Robustness to concurrent shocks: even at the conservative +2.8pp lower bound of
the 90% credible interval, a coincident shock would need to explain more than
+2.3pp of the measured lift before the no-decay annual value falls to the
€600,000 break-even (0.51pp) - wide headroom for the KEEP call.
```

Read-out. The verdict is one-sided in this run, and the table shows why: the measured **+3pp** lift is roughly six times the **0.51pp** break-even, so even a 3-month novelty half-life — which keeps only ~123 effective days of the 365 — still leaves ~ €1.2M of annual value against the €600k cost at P ~ 1.00 -> **KEEP**. The single exception is the most instructive row: under a harsh 1-month half-life the lift burns out before it pays (P ~ 0.00 -> *roll back/renew*). Notice what just happened to the decision: it no longer hinges on *whether* the redesign worked (it did — P(helped at all) > 0.99) but on **how long the lift persists** — a parameter the 60-day posterior cannot identify, because decay reveals itself only over a longer horizon. That's why the honest deliverable is the persistence *table* plus a monitoring plan (re-estimate the lift at day 120/180 and watch which row you're living in), not a single annualised number.

One number in the print block deserves top billing: the **realized** 60-day impact — the cumulative posterior converted to money — recovers most of the €600k build cost within the observed window alone, before any persistence assumption is invoked. Separating *realized* (a fact of the window we watched) from *projected* (a bet on the half-life) is the CMO-grade distinction the persistence table then builds on.

Memo to the CMO.

- The redesign works: P(lift > 0) > 0.99, and the measured lift is several times the ~0.5pp break-even.
- Sixty days in, the redesign has already paid back most of its €600k cost in *realized* conversion value (print above) — that part is banked, whatever novelty does next.
- Projected annual value clears the cost under every persistence scenario except a

1-month novelty burn-out: the open question is *how long the lift lasts*, not *whether it exists*. Re-estimate at day 120 and 180 and watch which table row we are living in.

- The estimate is only as good as the launch calendar: ITS books *anything* that changed on day 120 to the redesign (Section 5e). Before banking the number, audit the window for concurrent changes — pricing, campaigns, competitors, tracking.

0.6 7 · Caveats

- **The no-concurrent-shock assumption is the weak link.** A pricing change or competitor move at the redesign date gets attributed to the redesign — Section 5e shows the estimate absorbing an unrelated +2pp shock in full while every diagnostic stays green. Vet the calendar around the intervention; no statistic can do it for you.
- **Uncertainty grows with horizon** — the band (and cumulative-impact band) widens; long-horizon claims are genuinely less certain.
- **A control group beats extrapolation.** If even a small hold-out or comparable untreated series exists, DiD (nb 08) or synthetic control (nb 07) are stronger. ITS is the last resort when *everyone* was treated.
- **Autocorrelation.** If residuals are autocorrelated (checked above), a naive model understates uncertainty; prefer trend/seasonality structure or explicit AR terms.
- **Regression to the mean.** If the redesign shipped *because* conversion had dipped, the pre-period fit inherits the dip and the natural rebound is booked as lift. Ask what triggered the launch date; if the metric itself did, drop or model the dip window — and expect the placebo-date sweep (5d) to light up near it.